Year 11 Mathematics

Algebra 7

2.1 Using Formulae

Using Formulae

STARTER · 5 MIN

Date: _____ **Title:** Using Formulae

LEARNING OBJECTIVES

- Substitute values into formulae
- Use formulae with powers, roots and fractions
- Use formulae to solve problems

STARTER *Answer in your book. No calculators.*

1. Work out $3 \times (-4)$
2. Work out $(-5)^2$
3. Work out $2 \times 3^2 + 1$
4. Work out $\sqrt{16 + 9}$

Starter — Answers

STARTER

Tick or correct in green pen

1. $3 \times (-4) = \mathbf{-12}$
2. $(-5)^2 = \mathbf{25}$
3. $2 \times 9 + 1 = \mathbf{19}$
4. $\sqrt{25} = \mathbf{5}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can substitute values into formulae	☐
SILVER	I can use formulae with powers, roots and fractions	☐
GOLD	I can use formulae to solve problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Using Formulae

INTRODUCTION · 3 MIN

A formula shows the relationship between quantities. Substitute, then use BIDMAS.

Substitute

Replace each letter with its value.

$v = u + at$: $u = 3$, $a = 2$, $t = 5$

Brackets

Put negative numbers in brackets.

$x = -3$: $x^2 = (-3)^2 = 9$

Order

Brackets, indices, ×÷, then +−.

$v = 3 + 2 \times 5 = 13$

Units

Include units in the answer.

$A = 24\ \text{cm}^2$

Examples 1 and 2 — Substituting into Formulae

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

$v = u + at$

Find v when u = 4, a = 3 and t = 6.

Example 2

$P = 2(l + w)$

Find P when l = 7.5 and w = 3.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$v = 4 + 3 \times 6$

$v = 4 + 18$

$v = 22$

Tip: multiply before adding.

Example 2

$P = 2(7.5 + 3)$

$P = 2 \times 10.5$

$P = 21$

Tip: work out the bracket first.

Task 1

PRACTICE · 9 MIN

BRONZE

Answer in your book

1. $A = lw$. Find A when $l = 8$, $w = 5$

2. $y = mx + c$. Find y when $m = 3$, $x = 4$, $c = -2$

3. $s = d \div t$. Find s when $d = 150$, $t = 3$

4. $C = 2\pi r$. Find C when $r = 5$ (3 s.f.)

5. $T = 5n - 3$. Find T when $n = -2$

★ $F = 1.8C + 32$. Find F when $C = 25$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $A = 8 \times 5 = \mathbf{40}$

2. $y = 12 - 2 = \mathbf{10}$

3. $s = 150 \div 3 = \mathbf{50}$

4. $C = 2 \times \pi \times 5 = \mathbf{31.4}$

5. $T = -10 - 3 = \mathbf{-13}$

★ $F = 45 + 32 = \mathbf{77}$

Examples 3 and 4 — Powers, Roots and Fractions

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

$E = \frac{1}{2}mv^2$

Find E when m = 4 and v = −3.

Example 4

Find r (3 s.f.) when A = 50:

$r = \sqrt{\frac{A}{\pi}}$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$E = \frac{1}{2} \times 4 \times (-3)^2$$

$$E = 2 \times 9$$

$E = 18$

Tip: square v before multiplying.

Example 4

$$r = \sqrt{\frac{50}{\pi}}$$

$$r = \sqrt{15.915\ldots}$$

$r = 3.99$

Tip: work out the fraction, then the square root.

Task 2

PRACTICE · 9 MIN

SILVER

Give answers to 3 s.f. where needed

1. $s = ut + \frac{1}{2}at^2$. Find s when u = 2, t = 5, a = 4

2. $A = \pi r^2$. Find A when r = 6

3. Find T when l = 2: $T = 2\pi\sqrt{\frac{l}{10}}$

4. Find y when a = 6, b = −3: $y = \frac{a^2 - b}{2a}$

5. $V = \frac{1}{3}\pi r^2 h$. Find V when r = 3, h = 10

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $s = 10 + 50 = \mathbf{60}$

2 $A = \pi \times 36 = \mathbf{113}$

3 $T = 2\pi\sqrt{0.2} = \mathbf{2.81}$

4 $y = \dfrac{36 + 3}{12} = \mathbf{3.25}$

5 $V = \frac{1}{3} \times \pi \times 9 \times 10 = \mathbf{94.2}$

Examples 5 and 6 — Using Formulae Backwards

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5

(2 marks)

$v = u + at$

Find t when v = 30, u = 6 and a = 4.

Example 6

(3 marks)

$A = \frac{1}{2}(a + b)h$

Find h when A = 45, a = 7 and b = 8.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$30 = 6 + 4t$

$24 = 4t$

$\mathbf{t = 6}$

Tip: substitute, then solve the equation.

Example 6

$45 = \frac{1}{2} \times 15 \times h$

$45 = 7.5h$

$\mathbf{h = 6}$

Tip: simplify the known part first.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1. $P = 2l + 2w$. Find w when $P = 40$ and $l = 12$. (2 marks)

2. $v^2 = u^2 + 2as$. Find v when $u = 3$, $a = 2$, $s = 4$ ($v > 0$). (3 marks)

3. $F = 1.8C + 32$. Find C when $F = 104$. (2 marks)

4. $A = \pi r^2$. Find r when $A = 200$ (3 s.f.). (3 marks)

5. Taxi cost $C = 3 + 1.5d$. A journey costs £21. Find d. (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $40 = 24 + 2w, \quad w = \mathbf{8}$

2. $v^2 = 9 + 16 = 25, \quad v = \mathbf{5}$

3. $72 = 1.8C, \quad C = \mathbf{40}$

4. $r^2 = 200 \div \pi = 63.66, \quad r = \mathbf{7.98}$

5. $18 = 1.5d, \quad d = \mathbf{12\ km}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE $A = bh$. Find A when $b = 9$, $h = 4$

SILVER $y = 3x^2 - 2$. Find y when $x = -2$

GOLD $v = u + at$. Find a when $v = 25$, $u = 5$, $t = 4$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $A = 9 \times 4 = \mathbf{36}$

SILVER $y = 3 \times 4 - 2 = \mathbf{10}$

GOLD $20 = 4a, \quad a = \mathbf{5}$

Exam Practice — Substituting into Formulae

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

$v = u + at$

Work out the value of v when

$u = 5$, $a = -2$ and $t = 3$

v = ...

Question 2

(2 marks)

$T = 3n^2 - 5$

Work out the value of T when $n = -4$

T = ...

Exam Practice — Powers, Roots and Fractions

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(2 marks)

$$E = \frac{1}{2}mv^2$$

Work out E when m = 6 and v = 5

E = ...

Question 4

(2 marks)

$A = \pi r^2$

Work out A when r = 4.5

Give your answer to 3 s.f.

A = ...

Exam Practice — Using Formulae Backwards

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(3 marks)

$s = ut + \frac{1}{2}at^2$

Work out a when s = 50, u = 4 and t = 5

a = ...

Question 6

(4 marks)

$v^2 = u^2 + 2as$

(a) Work out v when u = 6, a = 2.5 and s = 12. Give your answer to 3 s.f. (2)

v = ...

(b) Work out s when v = 20, u = 8 and a = 4 (2)

s = ...

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$v = 5 + (-2) \times 3$ **M1**

$v = 5 - 6$

$v = -1$ **A1**

Tip: multiply before you add.

Question 2

$T = 3 \times (-4)^2 - 5$ **M1**

$T = 3 \times 16 - 5$

$T = 43$ **A1**

Tip: $(-4)^2 = 16$, not -16.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$E = \frac{1}{2} \times 6 \times 5^2$ **M1**

$E = 3 \times 25$

$E = 75$ **A1**

Tip: square v first.

Question 4

$A = \pi \times 4.5^2$ **M1**

$A = 63.617\ldots$

$A = 63.6$ **A1**

Tip: write the full answer before rounding.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$50 = 4 \times 5 + \frac{1}{2} \times a \times 5^2$ — M1

$50 = 20 + 12.5a$ — M1

$a = 2.4$ — A1

Tip: substitute, then solve the equation.

Question 6

(a) $v^2 = 36 + 60 = 96$ — M1

$v = \sqrt{96} = 9.80$ — A1

(b) $400 = 64 + 8s$ — M1

$s = 42$ — A1

Tip: square root at the end, not the start.